

that the business cycle is caused by real shocks and is not due to market failures or insufficient demand as in the models of John Maynard Keynes (1936). Real business cycle models have inflation, but inflation is *neutral* or has no real effects. These models are *production economies* because they involve optimizing firms producing physical goods, in addition to agents optimizing consumption and savings decisions, but the firms are subject to shocks that affect their output. One particularly important shock that affects firm output is a *productivity shock*. The early literature, like Kydland and Prescott (1982), did not have asset prices. (Kydland and Prescott won the Nobel Prize in 2004.) The next generation of models, like Jermann (1988), put them in. The newest papers, like Kaltenbrunner and Lochstoer (2010), capture realistic, and complicated, dynamics of shocks and agents' behavior.

Because these models are designed to work at business cycle frequencies, they are less relevant for investors who have short horizons. But for long-horizon investors—like certain pension funds, sovereign wealth funds, and family offices—the productivity factor should be considered. Asset returns reflect long-run productivity risk. At the time of writing, Europe is still in the throes of its sovereign debt convulsions, and an important issue is the future productive capacity of European economies. In Figure 7.5, I plot a five-year average of productivity shocks and stock returns. I use a five-year average because the productivity variable is used in economic models that are designed to explain business cycle variation, which has a frequency of three to six years. The productivity shocks are alternatively called *Solow residuals* after Robert Solow (1957) or *total factor productivity* (TFP) shocks. I take the TFP shocks constructed by Fernald (2009),

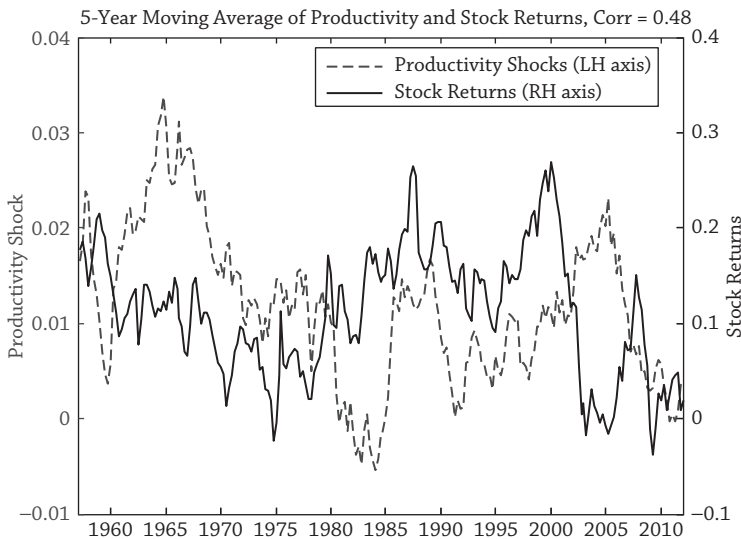


Figure 7.5